Genre Classification of Movies from Plot Descriptions

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Abstract

In this work, we investigate several approaches for predicting movie genres from plot descriptions using the TMDB Movies Dataset. We compare classical machine learning methods based on TF-IDF features with deep learning architectures including TextCNN, BiLSTM with Attention, and DistilBERT. To the best of our knowledge, no previously published benchmark results are available for this particular dataset split and experimental setup; therefore, the reported DistilBERT performance can be considered a strong baseline and the current state-of-the-art within this evaluation setting. Experimental results demonstrate the advantages and limitations of each approach and show the effectiveness of transformer-based models for text classification. Project code is available at: https://github.com/Blinov-Ilya/HSE_NLP_Project.git

1 Introduction

Modern streaming platforms and recommendation systems rely on automatic categorization of content to improve search quality, recommendation accuracy, and user experience. As media catalogs grow larger, accurate genre classification becomes increasingly important.

Movie plot descriptions contain rich semantic information about storylines, characters, and events, making them a valuable source for genre prediction. This makes genre classification from plot summaries a representative text classification task with practical applications in recommendation systems, content indexing, and information retrieval.

The goal of this work is to compare classical machine learning methods and modern transformer architectures for the task of movie genre classification.

The main contributions of this work are:

- analysis and preprocessing of the TMDB Movies Dataset (2021–2025), which contains recent movies and, to the best of our knowledge, has not been previously used as a benchmark for this genre classification task;
- implementation of multiple baseline approaches;
- comparative analysis of TF-IDF-based methods, recurrent and convolutional neural networks, and modern transformer architectures.

1.1 Team

Vsevolod Ershov prepared and preprocessed the dataset, implemented TF-IDF-based baseline models, and contributed to writing the report.

Ilya Blinov implemented and trained neural models and performed evaluation, including computation of all experimental metrics.

2 Related Work

Text classification is one of the most extensively studied tasks in Natural Language Processing (NLP). Early approaches relied on sparse vector representations such as Bag-of-Words and TF-IDF combined with linear classifiers including Logistic Regression and Support Vector Machines. Despite their simplicity, these methods remain strong baselines for many classification tasks due to their efficiency and interpretability [9].

The development of deep learning introduced neural architectures capable of learning contextual representations directly from text. Convolutional Neural Networks (CNNs) were successfully applied to sentence and document classification by extracting local semantic patterns through convolutional filters [1]. Recurrent architectures such as LSTM and BiLSTM further improved text classification by modeling sequential dependencies and long-range context within documents [2]. The addition of attention mechanisms enabled models to focus on the most informative parts of a text and significantly improved performance across various NLP tasks [3].

The introduction of Transformer-based architectures fundamentally changed the field of NLP. BERT demonstrated that large-scale bidirectional pretraining can substantially improve downstream text classification performance [4]. Later models such as DistilBERT [5] and RoBERTa [6] achieved competitive or superior results while improving training efficiency and representation quality.

Several studies have explored genre classification using movie plot descriptions. Ertugrul and Karagoz [7] proposed a Bidirectional LSTM-based approach for movie genre classification from plot summaries and demonstrated that BiLSTM models outperform traditional Logistic Regression and standard RNN baselines. Their work showed that sentence-level processing combined with prediction aggregation can improve classification performance on limited datasets.

More recent studies have focused on Transformer-based architectures. A recent work by Khan et al. [8] proposed an attention-based framework for multi-label movie genre classification and compared several Transformer models including BERT, DistilBERT, RoBERTa, ELECTRA, XLNet, and DeBERTa. Their results demonstrated the effectiveness of contextual language models for extracting genre-related information from plot descriptions.

This work focuses on the TMDB Movies Dataset (2021–2025), which contains more recent movie releases. To the best of our knowledge, no benchmark comparison of classical machine learning methods, neural architectures, and Transformer-based models has been reported on this dataset for the task of genre classification from plot descriptions.

3 Dataset

3.1 Source

In this work, we use the TMDB Movies Dataset (2021–2025), available on Kaggle: `https://www.kaggle.com/datasets/mechatronixs/tmdb-movies-dataset-20212025`.

The dataset contains **232,586 unique movie entries** identified by unique IDs and includes rich metadata such as movie titles, plot descriptions, genres, release dates, and additional auxiliary fields.

3.2 Task-Relevant Fields

For the purpose of this study, only textual and label-related fields are used:

- **overview** — plot description of the movie (main input text);
- **title** — movie title (optional, concatenated with overview);
- **genres** — list of genre labels (target variable);
- **release date** — used for temporal splitting (when available).

The final input representation is constructed as:

$$\text{text} = \text{title } [SEP] \text{ overview}$$

If the title is missing, only the overview is used.

3.3 Label Structure

The dataset originally contains **multi-label genre annotations**, where each movie may belong to multiple genres (e.g., Action, Adventure, Fantasy simultaneously).

Figure 1 shows the distribution of the top-15 most frequent genres in the dataset after preprocessing and filtering.

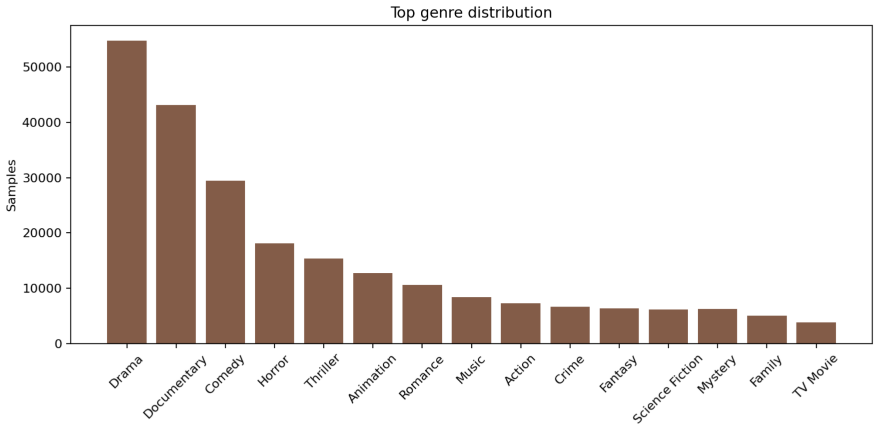

Figure 1: Distribution of the top-15 genres after preprocessing.

3.4 Preprocessing Pipeline

The dataset undergoes the following preprocessing steps:

1. removal of samples with missing or too short plot descriptions;
2. parsing and normalization of genre labels from heterogeneous formats;
3. removal of samples with empty genre lists after parsing;
4. construction of unified text field (title + overview).

Only samples with valid plot descriptions are retained. We further filter out entries where the overview length is below a predefined threshold.

3.5 Class Selection Strategy

Since the dataset contains a large number of unique genres, we restrict the label space to the **top-15 most frequent genres**, selected based on frequency in the training split.

All samples that do not contain at least one of the selected genres are removed.

This results in a controlled multi-label classification setting with a fixed label space:

$$y \in \{0, 1\}^{15}$$

3.6 Train/Validation/Test Split

We use a temporal split strategy: training on movies released up to 2023, validation on 2024, and testing on 2025.

3.7 Final Dataset Statistics

After applying the preprocessing pipeline described earlier, the resulting dataset consists of 155,031 unique samples. The data is split into training, validation, and test sets containing 87,575, 34,470, and 32,986 samples, respectively. The final dataset contains 15 unique genres.

4 Experiments

4.1 Metrics

Since movie genre classification is formulated as a multi-label classification problem, each movie may belong to multiple genres simultaneously. Therefore, standard single-label accuracy is not sufficient, and we use a set of multi-label evaluation metrics that capture different aspects of model performance.

4.1.1 Micro F1-score

Micro-averaged F1-score computes global precision and recall by aggregating contributions from all classes:

$$F1_{micro} = \frac{2 \cdot Precision_{micro} \cdot Recall_{micro}}{Precision_{micro} + Recall_{micro}}$$

This metric is sensitive to the overall number of correct predictions and is particularly suitable for imbalanced datasets, where some genres appear much more frequently than others. In this work, Micro-F1 is used as the primary evaluation metric.

4.1.2 Macro F1-score

Macro-averaged F1-score computes F1 independently for each class and then averages them:

$$F1_{macro} = \frac{1}{K} \sum_{k=1}^{K} F1_k$$

Unlike Micro-F1, Macro-F1 treats all genres equally, regardless of their frequency. This makes it useful for evaluating how well the model performs on rare genres, which are often underrepresented in the dataset.

4.1.3 Weighted F1-score

Weighted F1-score computes the F1-score for each genre independently and then averages the scores using class frequencies as weights:

$$F1_{weighted} = \sum_{k=1}^{K} \frac{n_k}{N} F1_k$$

where n_k denotes the number of samples containing genre k, N is the total number of label occurrences, and $F1_k$ is the F1-score for genre k.

Unlike Macro-F1, which assigns equal importance to all genres, Weighted F1 accounts for the class distribution in the dataset. As a result, performance on frequent genres contributes more to the final score than performance on rare genres.

4.1.4 Hamming Loss

Hamming Loss measures the fraction of incorrectly predicted labels:

$$HL = \frac{1}{N \cdot K} \sum_{i=1}^{N} \sum_{k=1}^{K} 1(y_{i,k} \neq \hat{y}_{i,k})$$

It evaluates errors at the level of individual labels rather than entire samples, making it a fine-grained metric for multi-label classification. Lower values indicate better performance.

4.1.5 Subset Accuracy

Subset Accuracy (also known as Exact Match Ratio) is a strict metric that requires the predicted set of labels to exactly match the ground truth:

$$Accuracy_{subset} = \frac{1}{N} \sum_{i=1}^{N} 1(Y_i = \hat{Y}_i)$$

This metric is very strict, as even a single incorrect label results in a zero score for the sample. However, it provides a clear measure of how often the model fully captures all genres correctly.

4.2 Most Frequent Classifier

As a trivial baseline, we use a most frequent class predictor, which always assigns the most common genre observed in the training set.

4.3 TF-IDF + Logistic Regression

This model represents a classical machine learning pipeline. Movie descriptions are first transformed into TF-IDF representations with up to 100,000 features and a minimum document frequency of 2. A Logistic Regression classifier is then trained using the scikit-learn implementation with a maximum of 1000 optimization iterations.

This approach assumes linear separability in the TF-IDF feature space.

4.4 TF-IDF + Linear SVM

In this model, TF-IDF features are used as input to a Linear Support Vector Machine classifier. The same vectorization settings are applied as in Logistic Regression. The model is optimized using a linear hinge-loss objective and is expected to perform well in high-dimensional sparse settings.

4.5 TextCNN

TextCNN is a convolutional neural network designed for text classification. The model consists of:

- token embedding layer with vocabulary size up to 50,000;
- convolutional filters over n-grams;
- global max-pooling layer;
- fully connected classification head with dropout (0.3).

Input sequences are truncated or padded to a maximum length of 256 tokens. The embedding dimension is set to 128.

4.6 BiLSTM with Attention

This model is based on a bidirectional LSTM encoder with hidden dimension 128. The architecture processes input sequences of up to 256 tokens using pretrained or learned embeddings of size 128.

Attention mechanism is applied over hidden states to compute a weighted representation of the sequence. The final representation is passed through a fully connected layer with dropout 0.3 for classification.

4.7 DistilBERT

DistilBERT is a transformer-based model obtained via knowledge distillation from BERT. We use the pretrained checkpoint `distilbert-base-uncased`.

The model is fine-tuned on the movie genre classification task using a maximum sequence length of 256 tokens and a batch size of 16. The learning rate is set to 2×10^{-5} with weight decay of 0.01.

Early stopping is applied with patience of 2 epochs based on validation performance.

4.8 Training Protocol

All neural and transformer-based models are trained using the same general pipeline:

- fixed random seed: 42;

- maximum number of epochs: 3;
- early stopping based on validation metric;
- batch size: 16;
- optimizer: AdamW (for deep models);
- learning rate scheduling depending on model type.

Classical models (TF-IDF + Logistic Regression and Linear SVM) are trained using scikit-learn with default optimizers and a maximum of 1000 iterations.

5 Results

Table 1 summarizes the performance of all evaluated models on the test set.

Table 1: Performance comparison of different models on the test dataset.

Model	**Macro-F1**	**Micro-F1**	**Weighted-F1**	**Hamming Loss**	**Subset Acc.**
Most Frequent (Baseline)	0.0363	0.2918	0.1302	0.1208	0.1695
TF-IDF + Logistic Regression	0.4708	0.5774	0.5797	0.0986	0.2792
TF-IDF + Linear SVM	0.4418	0.5511	0.5542	0.1041	0.2504
TextCNN	0.4025	0.5252	0.5246	0.1133	0.2478
BiLSTM + Attention	0.4481	0.5606	0.5659	0.1024	0.2938
DistilBERT	**0.5320**	**0.6318**	**0.6338**	**0.0822**	**0.3694**

The results demonstrate a clear performance hierarchy across the considered approaches. As expected, the Most Frequent baseline performs poorly on all metrics, confirming that genre prediction cannot be solved by exploiting only the marginal label distribution.

Among the classical machine learning approaches, TF-IDF combined with Logistic Regression achieves the strongest results, obtaining a Macro-F1 score of 0.471 and a Micro-F1 score of 0.577. Logistic Regression consistently outperforms Linear SVM on all reported metrics. This suggests that the linear decision boundaries learned by Logistic Regression are better suited to the sparse TF-IDF representation of movie plot descriptions.

The neural models improve upon the trivial baseline but do not consistently surpass the strongest TF-IDF baseline. BiLSTM with Attention achieves a Macro-F1 score of 0.448, while TextCNN reaches 0.403. Although both architectures are capable of modeling semantic information beyond bag-of-words representations, their performance remains below that of the TF-IDF + Logistic Regression baseline. This may be explained by the relatively short training schedule and the effectiveness of sparse lexical features for genre classification.

Transformer-based models achieve the best overall performance. DistilBERT obtains the highest scores across all F1 metrics, reaching a Macro-F1 of 0.532 and a Micro-F1 of 0.632. Compared to the strongest baseline (TF-IDF + Logistic Regression), DistilBERT improves Macro-F1 by approximately 0.061 and Micro-F1 by approximately 0.054. Furthermore, DistilBERT achieves the lowest Hamming Loss (0.082), indicating fewer label-level prediction errors.

The substantial improvement of DistilBERT over both classical and neural baselines highlights the importance of contextualized language representations for movie genre classification. Unlike TF-IDF-based approaches, transformer models can capture semantic relationships and long-range dependencies within plot descriptions, leading to more accurate genre predictions.

6 Conclusion

In this work, we investigated the problem of movie genre classification from plot descriptions using the TMDB Movies Dataset (2021–2025). The dataset contains recent movie releases and provides a realistic benchmark for evaluating modern text classification methods.

We compared a diverse set of approaches, including a Most Frequent baseline, TF-IDF-based linear models, neural architectures (TextCNN and BiLSTM with Attention), and transformer-based language models. The evaluation was performed using multiple metrics designed for multi-label classification, including Micro-F1, Macro-F1, Weighted-F1, Hamming Loss, and Subset Accuracy.

Experimental results show that classical machine learning methods remain competitive for this task. In particular, TF-IDF combined with Logistic Regression achieved the strongest performance among all non-transformer baselines. However, transformer-based models significantly outperformed both classical and neural approaches.

DistilBERT achieved the best overall performance, reaching a Macro-F1 score of 0.532 and a Micro-F1 score of 0.632. The improvement over the strongest baseline demonstrates the benefit of contextual language representations for understanding movie plot descriptions and identifying genre-specific patterns.

Overall, the results indicate that modern transformer architectures are currently the most effective solution for genre classification from textual movie summaries. At the same time, strong TF-IDF baselines remain valuable due to their simplicity, interpretability, and computational efficiency.

Future work may explore larger transformer models, improved threshold selection strategies for multi-label prediction, data augmentation techniques, and the incorporation of additional metadata.

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